AKSHAY JAITLY

Senior AI Engineer — Backend & AI-Powered Applications · Python, FastAPI, APIs, SQL
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SUMMARY

Senior backend engineer with 8+ years building production Python and Go APIs, microservices and data pipelines, now focused on AI-powered applications: integrating OpenAI and Google Gemini models into FastAPI services with RAG, agents and evaluation gates. Shipped fintech APIs at Capital One and real-time integration services at Siemens; owns work end to end, from API design and SQL data models to CI/CD and observability.

CORE SKILLS

- **Backend & APIs:** Python (FastAPI), Go, Node.js; RESTful, versioned APIs; microservices and service-oriented architecture; async and event-driven processing; integration layers for external and legacy systems
- **AI Integration:** OpenAI and Google Gemini APIs via a provider-agnostic LLM layer (also Groq, Ollama); RAG, hybrid retrieval, LangGraph and LangChain agents, tool use, prompt-injection defense, PII redaction, evals and AI observability
- **Data & SQL:** SQL, DynamoDB, Redis (cache-aside, connection pooling), Chroma vector store, ETL on AWS Glue and Spark, data reconciliation at 80M+ record scale
- **Cloud & DevOps:** AWS (Lambda, SNS, S3, DynamoDB, EC2, EMR, Glue, SageMaker), Docker, GitHub Actions CI/CD, IaC; learning Google Cloud (Vertex AI); pytest, Playwright

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Capital One · Chicago, IL · Jan 2025 – Present

- Design and build AI agents and RAG models in Python with LangChain and Langflow that automate multi-step decision-making across fintech business functions.
- Built reusable Claude AI skills adopted by 7 teams, packaging AI-assisted workflows as shared modules that standardized development and cut duplicated effort.
- Migrated 80M+ historical transactions from legacy systems and built the end-to-end ETL on AWS Glue 5, with API provisioning over the full dataset and automated daily reconciliation reports.
- Own CI/CD, infrastructure as code and observability end to end; containerized Python and Java 17 builds with Docker, reducing deployment failures.

Senior Software Engineer | Siemens Digital Industries Software · Chicago, IL · Aug 2022 – Jan 2025

- Built real-time REST APIs and connectors for a high-throughput IoT framework linking building systems across sites.
- Developed a semantic-enrichment engine and ML models in Python to enrich building and patient-room data.
- Designed Go REST APIs and Angular applications for Building X Sustainability Manager, Siemens' smart-building platform; improved dashboard load times by 15%.

Senior Software Engineer | Capital One · Aug 2021 – Aug 2022

- Built Go REST APIs on AWS Lambda for the Shop with Rewards point-of-sale product, enabling $15M in rewards redemption at REI in the first two months ($300M projected across partners); featured on the Capital One Tech blog.

Software Engineer | Capital One · Chicago, IL · Nov 2018 – Aug 2021

- Built event-driven automation with AWS Lambda, SNS and S3 triggers running transient EMR and EC2 clusters, cutting cloud operations cost 35%.
- Designed a scalable aggregation service handling bulk loads of up to 1M records; implemented cryptographic masking of PCI and NPI data across Spark, Scala and Spring Boot pipelines.

Earlier: HackerOne security researcher (2017–18) · SWE intern, Intl. Motor Group (2017) · MIT Pune research (2015–16)

SELECTED PROJECTS

Production RAG Agent · Python, FastAPI, LangGraph, OpenAI / Gemini / Groq, Chroma · github.com/AkshayJaitly/langchain-rag-assistant

- Shipped a FastAPI backend end to end with a provider-agnostic LLM layer (OpenAI, Google Gemini, Groq, Ollama) and a LangGraph pipeline: follow-up question rewriting, hybrid BM25 + vector retrieval, and a verifier agent that checks every claim and triggers a revision when it's unsupported.
- Hardened it for untrusted input: schema-validated requests, per-user document isolation, prompt-injection screening on uploaded documents, and secret and PII redaction on output.
- Built an evaluation harness that gates CI on retrieval quality; used it to decide on evidence, e.g. rejecting a multi-agent variant that doubled latency without improving accuracy.

redis-lab · Go, Redis, Docker · github.com/AkshayJaitly/redis-lab

- Built a Redis explorer with a hand-written RESP2/RESP3 protocol codec, a pooled client and a fail-closed command policy; its cache-aside demo cuts a ~370 ms origin call to ~1 ms on a hit.

EDUCATION

M.S., Computer Science · New Jersey Institute of Technology · 2018

B.E., Computer Engineering · University of Pune · 2016